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Two-Step Beamforming Scheme for Large-Dimension Reconfigurable Intelligent Surface

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Abstract—Beamformer design for large-dimension reconfigurable intelligent surface (RIS) is investigated. The challenge arises from the expanded near-field of RIS where its beamforming gain is, in general, distance-dependent due to near-field diffraction. To gain more insights into this effect, we first analyse the performance of conventional collimated beamforming using Fresnel integrals, and compare it with that of focused beamforming. Then, a two-step beamforming scheme is proposed. The principle of the proposed scheme is to divide the RIS into multiple subarrays followed by steering the sub-beams of the individual subarrays to focus. Providing properly chosen subarray size, the proposed scheme retains the low complexity of collimated beamforming and achieves a near-optimal performance, as shown both analytically and numerically.

Index Terms—RIS, beamforming, near-field, Fresnel integral, subarray

I. INTRODUCTION

Reconfigurable intelligent surface (RIS) has emerged as a promising new radio network topology technique to achieve extreme high data rate and extreme coverage extension of 6G [1]. In spite of being a passive array, RIS can manipulate the scattered waves as desired by adapting the scattering coefficients, e.g., the phase shifts, of the individual elements. For instance, RIS can be deployed in a cellular scenario to relay signals from a base station (BS) towards intended user equipments (UEs) for coverage enhancement, as depicted in Fig. 1. As compared to wireless relaying, a well-studied technique for manipulating the wireless channel, RIS is usually considered as a more economical alternative thanks to the properties such as simple driver circuits, no radio-frequency chains, and full-duplex operation mode [2], [3].

Due to its passiveness, RIS relies on beamforming to combat the high pathloss of the two-hop channel. Fortunately, RIS does not introduce significant noise. Therefore, the beamforming gain of RIS is quadratically proportional to its electrical size, which is also known as the “square law” [4]. Even though, in order to achieve a competitive gain with other radio network topology techniques¹, the required RIS size shall be markedly large. For a rough estimate, a $1\text{m} \times 1\text{m}$ RIS operating at 30 GHz provides a maximum beamforming gain of ca. 92 dB. However, such a large-dimension RIS has a near-field expanding up to 400 meters. In typical cellular scenarios, this

¹As a benchmark, the required end-to-end gain of mmWave smart repeater is approximately 90 dB [5].

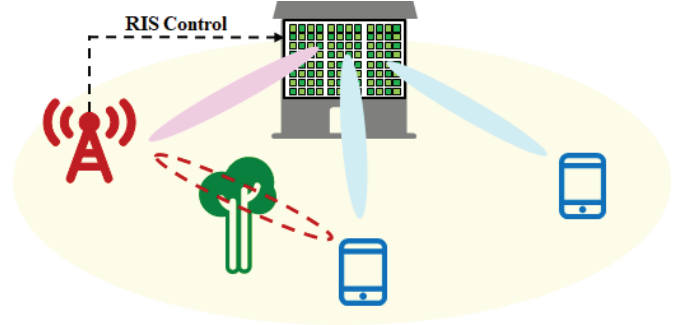


Fig. 1. RIS-aided cellular communication scenario

may cover the BSs and UEs it serves, especially the latter, in most cases. Hence, near-field effect has to be taken into account for the beamformer design of RIS, especially large-dimension ones.

In the near-field of RIS, the propagation of scattered waves is dominated by Fresnel diffraction. Thus, the beamforming gain of RIS is distance-dependent, in addition to the angles of the incident and scattered waves. As a result of this effect, the conventional collimated beamforming, which only depends on the directions and is optimal in the far-field, suffers from notable performance degradation in the near-field [6], [7]. More specifically, the effective beamforming gain has a limit which is independent of the RIS size or the distance between the RIS and the receiver. In other words, the promised “square law” fails. On the contrary, focused beamforming, as the near-field form of coherent beamformer, does not have such limitation. However, focused beamforming requires accurate adaptation of the focal point and yields higher implementation complexity.

This work is focused on the beamformer design for large-dimension RIS in a typical line-of-sight (LOS) scenario. To gain more insights into the near-field effect, the pathloss of collimated beamforming will be studied using Fresnel integrals, which yields an analytically tractable expression providing paraxial conditions. Based on the analysis, a two-step RIS beamforming scheme is proposed aiming at a good performance-complexity trade-off. The principle of the proposed scheme is to first divide RIS into multiple subarrays according to the distance to the user. Then, the sub-beams

generated by the individual subarrays are steered to focus on the UE, in order to approach the performance of focused beamforming with much lower complexity. Analysis and numerical simulations show that the proposed scheme achieves a near-optimal performance in the near-field.

In the next section, a free-space pathloss model from literature is adopted as a basis of our discussion. Focused and collimated beamforming and their performances are discussed in Section III. The proposed beamforming scheme is introduced afterwards. Finally, numerical simulation results are provided in Section VI to demonstrate the performance of the proposed scheme in more practical channels.

II. SYSTEM MODELLING

An RIS-aided cellular scenario is considered, as depicted in Fig. 1, where a single RIS is deployed between a BS and a UE to relay the downlink transmission. Both BS-RIS and RIS-UE links are assumed to be LOS. A single data stream is to be transmitted over the BS-RIS-UE link. For the considered transmission strategy, the existence of the BS-UE direct link does not influence the design of the RIS beamforming scheme and, hence, will be omitted in this paper.

The modelling of RIS channel consists of the modelling of the RIS itself and the modelling of the physical channels, both of which are in general on going research topics [6], [8]. In this paper, the free-space pathloss model proposed in [8] is tailored to explicitly model the RIS-UE link. Specifically, we consider a uniform rectangular RIS of size $L_x \times L_y$ consisting of $M \times N$ elements. Let the UE receive signal power P_r be written as

$$P_r = \frac{G}{L(d)} \left| \sum_{m=1}^M \sum_{n=1}^N \sqrt{P_{m,n}} e^{j\phi_{m,n}} e^{-j\frac{2\pi}{\lambda} |\vec{r} - \vec{r}_{m,n}|} \right|^2, \quad (1)$$

where the notations are defined in the following.

G : maximum reradiation gain of RIS element

$L(\cdot)$: free-space pathloss between isotropic antennas

$P_{m,n}$ and $\phi_{m,n}$: reradiated power and phase angle of a single RIS element, respectively

$\vec{r}$ and $\vec{r}_{m,n}$: vectors from RIS center to the UE and to a RIS element in Cartesian coordinate system, respectively

$d = |\vec{r}|$: distance from RIS center to the UE

λ : the wavelength

In (1), the UE receive antenna is simply assumed to be isotropic with unit gain, without loss of generality. The maximum reradiation gain G of RIS element can be given by

$$G = \epsilon \frac{4\pi}{\lambda^2} \Delta x \Delta y, \quad (2)$$

where ϵ denotes the efficiency, and $\Delta x = L_x/M$ and $\Delta y = L_y/N$ denote element spacing. In this paper, the efficiency ϵ is assumed to be one unless specified otherwise.

The free-space pathloss $L(d)$ is computed as

$$L(d) = \left(\frac{4\pi d}{\lambda} \right)^2. \quad (3)$$

As long as the distance between RIS and UE is much larger than the size of RIS, it is valid to neglect the impact of small distance variations between individual RIS elements and the UE on the pathloss.

The reradiated power $P_{m,n}$ absorbs the transmit power and the antenna gain of BS, the pathloss between BS and RIS, and the power pattern of RIS element in the direction of the UE in [8]. Specially, we define the effective sum power reradiated by RIS in the direction of the UE as

$$P_s = \sum_{m=1}^M \sum_{n=1}^N P_{m,n}. \quad (4)$$

The reradiated phase angle $\phi_{m,n}$ absorbs the imping signal phase and the phase shift applied by RIS elements. Furthermore, it is assumed, ideally, that the reradiated signal phase angles $\phi_{m,n}$ can be independently and accurately tuned by RIS elements.

III. FOCUSED AND COLLIMATED BEAMFORMING

To maximize the UE receive signal power P_r in (1), the optimum RIS reradiated phase angle $\phi_{m,n}$ is known to be

$$\phi_{m,n}^{\text{focused}} = \frac{2\pi}{\lambda} |\vec{r} - \vec{r}_{m,n}|, \quad (5)$$

and the corresponding UE receive signal power P_r is

$$P_r^{\text{focused}} = \frac{G}{L(d)} \left| \sum_{m=1}^M \sum_{n=1}^N \sqrt{P_{m,n}} \right|^2, \quad (6)$$

which is maximized when the reradiated signal power $P_{m,n}$ is evenly distributed over RIS elements². In the context of RIS beamforming, this scheme is usually referred to as focused beamforming.

Now, consider a normalizing distance

$$d_R = \frac{MN\Delta x\Delta y}{\lambda} = \frac{L_x L_y}{\lambda}. \quad (7)$$

The above distance d_R is not the usually considered far-field boundary, i.e., the Fraunhofer distance. For a square RIS, d_R equals 1/4 of the Fraunhofer distance, and is sometimes referred to as the Rayleigh distance. Furthermore, the maximum of the UE receive signal power in (6) can be reformulated as

$$P_r^{\text{focused}} = \frac{\lambda^2 P_s}{4\pi L_x L_y} \cdot \frac{d_R^2}{d^2}. \quad (8)$$

The above expression suggests that a UE located at the normalizing distance d_R receives exactly the same power as if it would be located at the position of the RIS. That is to say, the pathloss of the RIS-UE link is fully compensated by the focused beamforming gain of RIS.

In contrast to focused beamforming, collimated beamforming adopts the far-field condition and assumes that the rays

²Note that this requires a reversed power tapering if non-isotropic power radiation patterns of RIS elements are considered.

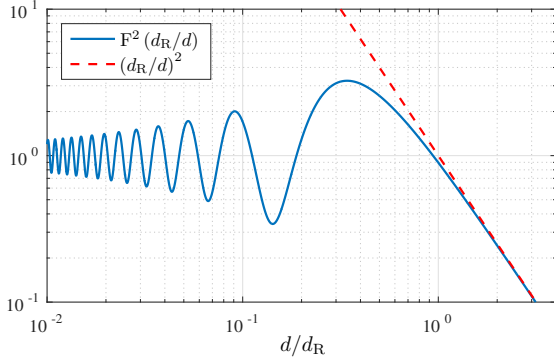


Fig. 2. Comparison of $F^2(d_R/d)$ and $(d_R/d)^2$, representing the axial performance of collimated and focused beamforming in the near-field of square RIS, respectively.

from RIS elements to the UE are parallel. Therefore, the square-root distance in (5) simplifies to

$$|\vec{r} - \vec{r}_{m,n}| \approx d - \frac{1}{d} \vec{r}_{m,n} \cdot \vec{r}, \quad \text{for } d \gg d_R. \quad (9)$$

The corresponding RIS reradiated phase angle reads

$$\phi_{m,n}^{\text{collimated}} = -\frac{2\pi}{\lambda d} \vec{r}_{m,n} \cdot \vec{r}. \quad (10)$$

However, this simplification leads to a non-negligible phase mismatch with the channel in the near-field and, hence, a notable performance degradation. This will be analyzed in the following considering a UE being located in the axial direction of a rectangular RIS with evenly distributed reradiated power. In this case, a more accurate approximation to the square-root distance in (5) is the well-known Fresnel approximation

$$|\vec{r} - \vec{r}_{m,n}| \approx d + \frac{1}{2d} |\vec{r}_{m,n}|^2, \quad \text{for } d \sim d_R. \quad (11)$$

If collimated beamforming is exploited, the UE receive power P_r can be approximated using (11) as

$$P_r^{\text{collimated}} \approx \frac{\lambda^2 P_s}{4\pi L_x L_y} \cdot F\left(\frac{L_x^2}{\lambda d}\right) F\left(\frac{L_y^2}{\lambda d}\right), \quad (12)$$

with

$$F(x) = 2 \left[C^2\left(\sqrt{x/2}\right) + S^2\left(\sqrt{x/2}\right) \right] \quad (13)$$

and

$$C(x) = \int_0^x \cos\left(\frac{\pi t^2}{2}\right) dt \quad \text{and} \quad S(x) = \int_0^x \sin\left(\frac{\pi t^2}{2}\right) dt. \quad (14)$$

The function $F(x)$ in (13) is introduced for notational simplicity, and $C(x)$ and $S(x)$ are Fresnel integrals following the definition in [9, Sec. 7.2(iii)]. For a square RIS with $L_x^2/\lambda = L_y^2/\lambda = d_R$, (12) reduces to

$$P_r^{\text{collimated}} = \frac{\lambda^2 P_s}{4\pi L_x L_y} \cdot F^2\left(\frac{d_R}{d}\right). \quad (15)$$

The derivation of (12) to (15) is omitted in this paper, but a similar formula to the square case (15) has been given in [10].

As illustrated in Fig. 2, the function $F(\cdot)$ has a damped oscillating behavior around one as d reduces, following the convergence behavior of the Fresnel integrals (14). Deduced from this, the limitations of collimated beamforming in the near-field can be viewed from the following two aspects. Firstly, the pathloss is oscillating within the range of d_R , and the average value approximately equals that at the distance of d_R . Secondly, increasing the RIS size only extends the range of d_R , but does not improve the maximum gain. This suggests that the “square law” fails in the near-field if collimated beamforming is used.

IV. PROPOSED BEAMFORMING METHOD

To overcome the above limitations of collimated beamforming in the near-field while retaining its simplicity, a two-step beamforming scheme is proposed for large-dimension RIS. Let the distance from a RIS element to the UE be approximated as

$$|\vec{r} - \vec{r}_{m,n}| \approx d - \frac{\vec{r}_{m,n} \cdot \vec{r}}{d} + \frac{|\vec{r}_{m,n}|^2}{2d} \quad (16)$$

by omitting the second and higher order terms in the binomial expansion of $\sqrt{1+x}$. Equation (16) essentially describes another method to approach the near-field performance of focused beamforming. However, we would like to further eliminate the quadratic term $|\vec{r}_{m,n}|^2$ in (16).

Let the RIS be divided into $I \times J$ rectangular non-overlapping subarrays. In general, the subarrays can be of different size. Suppose the (m, n) element belongs to the (i, j) subarray. Thus $\vec{r}_{m,n} = \vec{q}_{m,n} + \vec{s}_{i,j}$ holds, where $\vec{s}_{i,j}$ represents the reference point of the subarray and $\vec{q}_{m,n}$ represents the relative position of the element in the subarray. Equipped with this, the quadratic term in (16) can be relaxed as

$$\frac{|\vec{r}_{m,n}|^2}{2d} \approx \frac{\vec{q}_{m,n} \cdot \vec{s}_{i,j}}{d} + \frac{|\vec{s}_{i,j}|^2}{2d}, \quad (17)$$

where the quadratic term of $|\vec{q}_{m,n}|$ is omitted. Combining (16) and (17) yields the reradiated phase angle of RIS elements for the proposed beamformer

$$\phi_{m,n}^{\text{proposed}} = -\frac{2\pi}{\lambda d} \vec{r}_{m,n} \cdot \vec{r} + \frac{2\pi}{\lambda d} \vec{q}_{m,n} \cdot \vec{s}_{i,j} + \frac{\pi}{\lambda d} |\vec{s}_{i,j}|^2. \quad (18)$$

In (18), the first and the second term depends on the angle and the distance, respectively, to the UE, and the third term is a common phase shift applied for all elements of a subarray. Thus, the proposed beamforming method yields a constant gradient in each individual subarray.

The proposed beamforming method can be interpreted as follows. In a first step, each RIS subarray generates a collimated sub-beam in a common direction $\vec{r}/d$ towards the UE. This corresponds to the first term in (18). Then, each RIS subarray steers its sub-beam by an angle specified by the position of the subarray and the distance to the UE, corresponding to the second term in (18). This ensures that all sub-beams are focused on the UE. Finally, each subarray applies an initial phase shift such that the receive signals add coherently, corresponding to the third term in (18). As

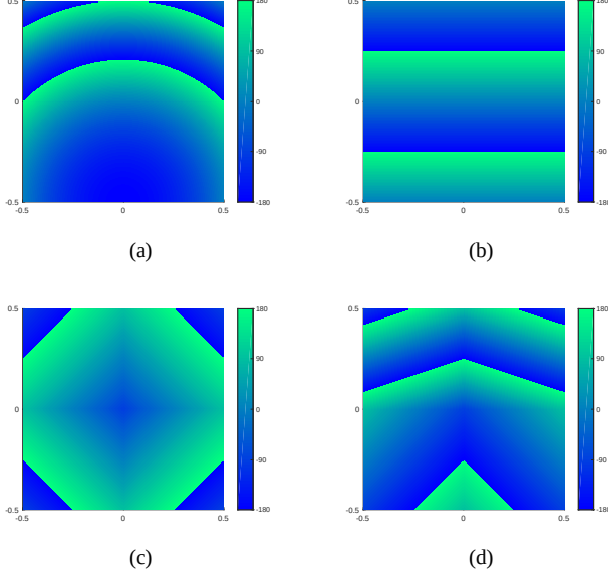


Fig. 3. An example of reradiated phase angles $\phi_{m,n}$ for: (a) focused beamforming in (5), (b) collimated beamforming in (10), (c) the second and third terms in (18), and (d) proposed method in (18). The RIS-UE distance is $d_R/5$.

an example, the reradiated phase angles of RIS elements for focused, collimated, and the proposed beamforming are illustrated in Fig. 3 for a square RIS with 4 subarrays.

The near-field performance of the proposed beamforming method is influenced by the choice of the subarray size. This will be discussed in the following. Consider a UE in the axial direction of a rectangular RIS with evenly distributed reradiated power, as in Section III. If the proposed beamforming method is exploited and the RIS is divided into $I \times J$ subarrays of common size, the UE receive power can be computed as

$$P_r^{\text{proposed}} = \frac{I^2 J^2 \lambda^2 P_s}{4\pi L_x L_y} \cdot F\left(\frac{L_x^2}{I^2 \lambda d}\right) F\left(\frac{L_y^2}{J^2 \lambda d}\right). \quad (19)$$

Comparing (19) with the axial performance of collimated beamforming given in (12), the following two advantages can be concluded. Firstly, the near-field performance of the proposed method can be improved up to a factor of $I^2 J^2$. The promised “square-law” gain of large-dimension RIS can be attained in the near-field. Secondly, the fine pathloss behavior in the near-field now depends on the size of subarrays, i.e., L_x/I and L_y/J . The damped oscillating behavior appears at a distance which is reduced by a factor of approximately IJ . Based on (19), one could divide a large-dimension RIS into more subarrays of smaller size in order to maximize the performance. However, according to the fine behavior of Fresnel integrals demonstrated in Section III, it shall be sufficient to divide the subarrays such that the distances $L_x^2/I^2 \lambda$ and $L_y^2/J^2 \lambda$ are comparable to the RIS-UE distance d . This achieves a reasonable performance-complexity trade-off in the near-field of RIS.

V. BS-SIDE PHASE COMPENSATION

So far, only the RIS-UE link has been considered. The BS-RIS link will be discussed briefly in this section. One major difference between the BS-RIS link and the RIS-UE link is that BSs are typically equipped with more antennas than a UE, resulting in more degrees of freedom in the joint optimization of BS-RIS beamforming. However, the far-field of a mmWave BS is typically a few meters. As long as the RIS is deployed in the far-field of the BS, it is reasonable for the BS to beamform towards the RIS. In this case, the incident wave at RIS is approximately a spherical wave with almost constant power density. Hence, the RIS beamforming problem can be decoupled into two subproblems: 1) the RIS-UE beamforming as discussed in the previous section, and 2) the BS-side phase compensation.

If the RIS is deployed at a fixed location, the BS-side phase compensation can be designed a priori. If mobility of RIS is considered, the proposed beamforming method can also be directly adopted to this end. It only needs to be noticed that the subarray size for BS-side phase compensation shall be adjusted according to the BS-RIS distance, which is not necessarily the same as that for RIS-UE beamforming.

Alternatively, the compound RIS phase shift could be design in one step by jointly considering the BS-RIS side and the RIS-UE side beamforming. Examples of such approach include [11], [12], where the concept of Fresnel region is exploited. These approaches are based on a similar principle as the proposed beamforming scheme.

VI. SIMULATION RESULTS

The coverage performance of the proposed RIS beamforming scheme will be demonstrated using the following numerical simulations. Consider a $1 \text{ m} \times 1 \text{ m}$ square RIS operating at 30 GHz. The BS antennas are assumed to be vertically polarized. A Rician fading channel with $K = 10$ is assumed for the BS-RIS link. Specially, the classic two-ray ground reflection channel is adopted to model the RIS-UE link. Besides, the BS-UE direct link is assumed to be unavailable. Other parameters are listed in Table I.

In Fig. 4, the receive SNR at the UE is plotted as a function of the RIS-UE horizontal distance. In this simulation, the shadow fading of the BS-RIS link is omitted to better illustrate the near-field behavior of different beamforming schemes. The previously discussed focused and collimated beamforming as well as the optimum fully coherent beamforming are compared with the proposed beamforming scheme. Firstly, the typical two-ray fading pattern vanishes in this case, mainly due to the “channel hardening” effect. Secondly, the near-field performances of focused, collimated, and the proposed beamforming match the analytical results quite well. A damped oscillating behavior can be observed if the RIS-UE distance is smaller than d_R , except for very small distances where Fresnel approximation is no longer valid. For the proposed beamforming scheme with 2×2 and 4×4 subarrays, the gain w.r.t. collimated beamforming is up to 12 dB and 24 dB in the near-field, as revealed by (19).

TABLE I
SIMULATION PARAMETERS

Parameter	Value
Carrier frequency	30 GHz
Bandwidth	400 MHz
BS Tx power	27 dBm
BS antenna array	8×8
BS antenna max. gain	8 dBi (vertical polarization) [13]
BS height	10 m
BS-RIS distance	150 m
RIS size	$1 \text{ m} \times 1 \text{ m}$
RIS element spacing	$\lambda/2$
RIS element gain	2 dBi
RIS height	2 m
UE antenna gain	0 dBi (isotropic)
UE noise figure	10 dB
UE height	1.5 m
BS-RIS pathloss	UMi-Street Canyon LOS [13]
Shadowing	$\sigma = 4$
Other loss	10 dB

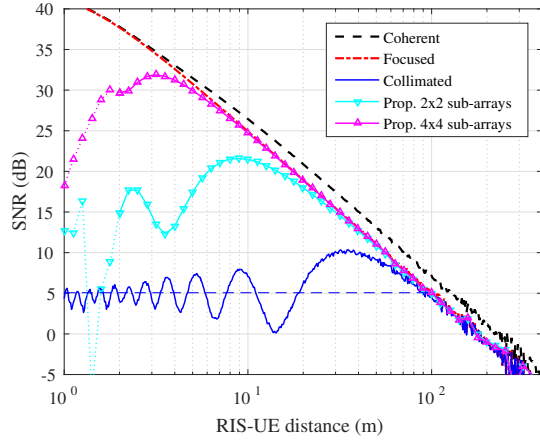
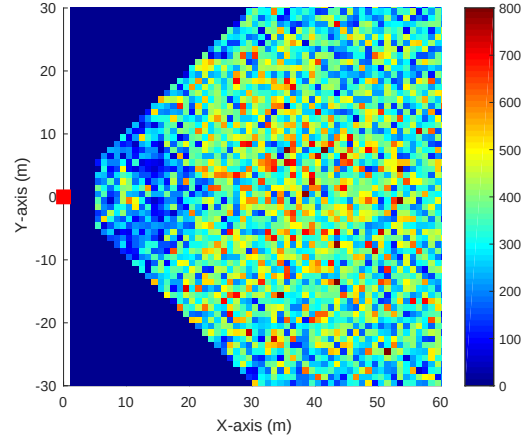


Fig. 4. UE receive SNR in the near-field of a large-dimension RIS with two-ray channel model

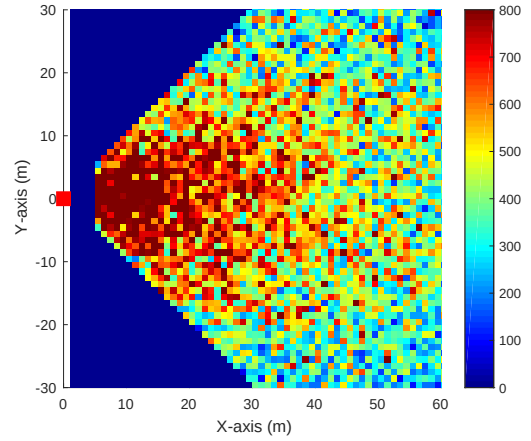
In Fig. 5, the coverage in terms of achievable throughput is evaluated for collimated and proposed beamforming, respectively. In this case, shadow fading is considered. On average, the proposed scheme notably outperforms collimated beamforming within a range of ca. 50 m. It shall be noticed that for collimated beamforming, higher throughput can be achieved at the distances of ca. 10 m and 30 m, corresponding to the peaks of Fresnel integrals. For the proposed beamforming scheme, the performance degrades with distance almost monotonically.

VII. CONCLUSIONS

In this paper, we tackle the problem of beamformer design for large-dimension RIS with a focus on the near-field. It is shown that the performance limit of conventional beamforming in the near-field can be well described by Fresnel integrals. Based on the analysis, we propose to approach the optimal performance by generating focused sub-beams from subarrays of RIS. The proposed scheme inherits simplicity from conven-



(a)



(b)

Fig. 5. Achievable throughput in Mbps with (a) collimated beamforming and (b) the proposed beamforming exploiting 4×4 subarrays. The RIS is represented by the square at origin.

tional beamforming and achieves a near-optimal performance as shown by simulated results. If cross-polarization is further exploited, it is promising to provide a “Gbps everywhere” coverage with the help of RIS.

A practical challenge of large-dimension RIS is the individual control of massive elements. To tackle this, the proposed scheme can also be extended. For instance, the subarrays can be divided even smaller and a common phase shift is applied to all RIS elements in the subarray. In this case, the performance can be analyzed in a similar way as done in this paper. Beside this, other practical challenges such as phase quantization of RIS elements and the inaccuracy in position estimation have not been addressed yet. These topics will be left for future work.

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